

Endong Dai

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PROFILE

ECE M.S. candidate at Duke specializing in RTL design & verification for AI accelerators. Architected an LLM accelerator's Matrix Multiplication Unit (systolic GEMM/GEMV, ~1.0 TOPS, TSMC 16nm) and helped drive a taped-out diffusion SoC through full-model silicon validation.

EDUCATION

Duke University	M.S. in ECE -Hardware Track (GPA 3.8/4.0 First Year)	Aug 2025 – May 2027
University of Nottingham	B.Eng. in EEE, First Class (GPA 3.8/4.0)	Sep 2021 – Jul 2025

TECHNICAL SKILLS

- **IC & FPGA Design:** Verilog, Vivado, Quartus Prime, ModelSim, FPGA (Xilinx ZCU102), Cadence Virtuoso
- **PCB & Simulation:** Altium Designer, KiCad, LTspice, PLECS, MATLAB/Simulink, Soldering
- **Programming & Embedded:** C/C++, Python, Assembly, STM32CubeIDE, Arduino, Raspberry Pi, ROS
- **Tools & Platforms:** Claude Code, Linux(Ubuntu), Git

PROJECT EXPERIENCE

LLM Accelerator RTL Design – Matrix Multiplication Unit (MMU) Jan 2026 – Jun 2026

RTL Design Engineer – Duke Center of Computational Evolutionary Intelligence (CEI)

- Architected the Matrix Multiplication Unit (MMU), an LLM accelerator's central GEMM/GEMV engine; a K-means codebook quantization recasts memory-bound decode GEMV as dense GEMM for a **10× decode speedup**.
- Implemented two native operating modes on a shared PE array and control path: full-precision prefill GEMM and codebook-indexed quantized GEMV for optimized decode.
- Designed a **32×32 INT8 systolic PE array (1,024 MACs)** with AXI-Stream ready/valid dataflow delivering **~1.0 TOPS**, verified against software golden models and synthesized in a **TSMC 16 nm** flow.

Diffusion Chip – SoC Bring-Up

Aug 2025 – Jul 2026

SoC Bring-up Engineer – Duke CEI

- Drove FPGA validation and post-silicon bring-up of a taped-out diffusion-accelerator SoC (**TSMC 16nm, CPGA-208**; RISC-V CPU + NPU, **1 MB GLB SRAM**, dual **32-bit FX3 USB** on a shared AXI fabric; **INT8 1.024 TOPS** / FP16 0.256 TOPS), first on an **AMD ZCU102** FPGA prototype then on real silicon.
- Brought up the full diffusion UNet on real silicon — all **762 hardware-scheduled nodes** of an 839-node ONNX graph — with bit-exact **0-mismatch** against the scheduler/C-model golden across **41K+ dumped node tensors**.
- Root-caused and fixed 12 independent bring-up bugs across firmware operator scheduling, NPU hardware modes, and FX3 USB packet/ACK protocol, advancing the on-chip run from node n9 to full-model completion (n762).

CPGA-208 Diffusion Chip Carrier PCB Design

Feb 2026 – Jun 2026

PCB Project Engineer – Duke CEI

- Led chip bonding and **PCB** design of a **CPGA-208** carrier board for the diffusion chip on a 6-layer **Altium Designer** platform, including **length matching** for **dual 32-bit** high-speed FX interfaces.
- Designed **multi-domain power** with LDO-based regulation, level shifting and power filtering networks.

Cadence Virtuoso IC Design (TSMC 65nm) – 16-bit ALU & PLL FM Receiver

Sep 2025 – Apr 2026

Digital / Analog-Mixed-Signal IC Design Engineer (Course Project) – Duke University

- Designed schematic and full-custom layout of an **IEEE-754 half-precision ALU** (16 MHz, <18 mW) and a PLL-based multi-channel FM receiver (VCO, phase detector, charge pump).

TEACHING EXPERIENCE

Teaching Assistant – Duke University

Summer – Fall 2026

Department of Electrical & Computer Engineering

- Course TA for *ECE550 Fundamentals of Computer Systems and Engineering* (Fall 2026): held office hours, graded assignments, and mentored students on RTL design and verification.